

Alternative: plain switches + Tailscale on generic Linux, no GL-iNet routers

Same theatre topology (isolated /24 per theatre, subnet routing, Tailscale ACLs doing cross-theatre isolation) — different hardware. Instead of a GL-iNet A-1300 doing router+NAT+DHCP+Tailscale in one appliance, use a plain unmanaged switch for LAN fanout plus a small generic Linux box (mini-PC/NUC, or the theatre's existing control laptop) running Tailscale as a subnet router with NAT/DHCP built from standard Linux tools (nftables + dnsmasq). Config in config/alt-tailscale-switch/.

Why a router-equivalent device is still required, not eliminated: the ATEM and BirdDog Play are closed appliances — they can't run the Tailscale client themselves. Some device must still advertise the theatre's subnet on their behalf. "Pure Tailscale, no router" only works if literally every device can run Tailscale; since two can't, this alternative just moves the subnet-router role onto generic Linux instead of GL-iNet's firmware. The tailscale up invocations themselves are identical either way — see config/tailscale-up-all-devices.sh — Tailscale doesn't care what hardware runs it.

Comparison

	GL-iNet A-1300 (chosen)	Generic Linux box + switch (alternative)
Setup effort	Config file, done (see config/gl-inet/)	Build + harden NAT/DHCP/Tailscale from scratch per box
Throughput confidence	Vendor datasheet, ~170 Mbps — but that is kernel WireGuard; Tailscale's userspace wireguard-go is unbenchmarked on the A-1300 and likely slower (see docs/bandwidth-analysis.md)	Unverified until benchmarked on whatever's chosen (an x86 mini-PC will very likely beat the A-1300 under Tailscale)
Physical footprint	One compact unit per theatre	Separate PC + separate switch + 2 power supplies
Dedicated ATEM port	Built-in (2 LAN ports)	Needs a 2nd NIC/dongle to replicate
Wi-Fi	Built-in	Extra hardware if needed at all
Spares/swap-in	Re-flash identical UCI config	Re-provision a whole OS image
Maintenance	Vendor firmware updates	Ongoing OS patching, our responsibility
Flexibility/CPU headroom	Fixed, embedded-class	Higher, if it matters later
Unit cost at required throughput	~\$100-class travel router	Often more, once a real NIC + case + PSU are added

Recommendation: GL-iNet

This is 12 identical small router/switch appliances for a touring event that need to just work — exactly the product category a travel router is built for. A generic Linux box gives more flexibility and CPU headroom

we don't need here, at the cost of more setup work, more failure modes, and a bulkier kit per theatre. (The A-1300's own throughput "guarantee" is weaker than it looks under Tailscale — if the benchmark in `docs/bandwidth-analysis.md` comes back low, CPU headroom stops being something "we don't need" and this comparison should be revisited.) The one real advantage of the alternative — no vendor lock-in — doesn't outweigh those costs for this deployment. Keep GL-iNet as the primary; this alternative is documented for completeness, not as an equally-weighted option.